RES POLYTECHNIC, BANGALORE-91

MODEL ANSWER BOOKLET

CIE Theory Test I

Program	Computer Science and Engineering
Semester	III
Course Name	System and Network Administration
Course Code	25CS34I
Test	CIE Theory Test I (Duration 90 min, Max Marks 50)
Course Coordinator	Vinay Kumar
Coverage	Section I: Q1, Q2 (Hardware, PSU, SMPS, ESD) | Section II: Q3, Q4 (Chipset, Task handling, Boot process, POST, Performance, SSD/HDD)

Note to the student

- In the exam: answer **any ONE full question from each section** (Q1 or Q2; Q3 or Q4). Each full question carries 25 marks.
- This booklet solves **both alternatives** of every question for practice.
- Marks split: part (a) = 5, part (b) = 10, part (c) = 10. Write points and draw the diagrams neatly with labels.
- Beep-code meanings depend on the BIOS manufacturer; this is stated clearly in Q4(a).

SECTION I

Q.1 (a) [5 Marks]

Question: *Classify the given devices/components into suitable categories (input, output, storage, memory, networking).*

Sl.	Device / Component	Category	Reason
i	Keyboard	Input device	Sends typed data/commands to the computer
ii	Smartwatch Screen	Output device (also input as touchscreen)	Displays time, notifications; touch gives input
iii	ROM	Memory (primary, non-volatile)	Stores BIOS/firmware permanently; content kept without power
iv	Ethernet Cable	Networking	Physical medium (twisted pair, RJ-45) for wired LAN
v	Memory Card	Storage (secondary, flash)	Portable non-volatile storage, e.g. SD card
vi	Wi-Fi Adapter	Networking	Provides wireless connection to a network
vii	Plotter	Output device	Prints large drawings/vector graphics using pens
viii	Webcam	Input device	Captures video/images and sends to computer
ix	Register	Memory (CPU internal, fastest)	Small, very fast storage inside CPU for data/instructions

Summary

Input	Output	Memory	Storage	Networking
Keyboard, Webcam	Smartwatch Screen, Plotter	ROM, Register	Memory Card	Ethernet Cable, Wi-Fi Adapter

Q.1 (b) [10 Marks]

Question: *Kiran's uncle wants to set up a computer for his small business. What hardware would you suggest and why?*

Step 1 - Understand the requirement: A small business normally needs billing/accounting software, MS Office, internet and email, printing and data backup. It does not need heavy gaming/graphics hardware. So a reliable, **cost-effective office desktop** is suitable.

Recommended hardware

Component	Suggestion	Reason (why)
Processor (CPU)	Intel Core i3/i5 or AMD Ryzen 3/5 (4 cores or more)	Smooth multitasking for accounting software, browser and Office at moderate cost
RAM	8 GB DDR4 (expandable to 16 GB)	Avoids slowdown when many programs are open; room for future growth
Storage	256/512 GB SSD (+ 1 TB HDD or external drive for backup)	SSD gives fast boot and quick software loading; HDD/external drive stores backups cheaply
Motherboard	Branded board compatible with CPU/RAM, with USB, SATA, LAN, Wi-Fi support	Stability and upgrade options
Graphics	Integrated graphics	Enough for office work; saves cost
Power supply (SMPS)	450 W branded ATX SMPS	Stable power protects other components
Monitor	21-22 inch Full HD LED	Comfortable viewing, low power use

Component	Suggestion	Reason (why)
Keyboard and Mouse	Standard USB/wireless	Basic input
Printer / Scanner	Laser multi-function printer	Low cost per page for invoices and bills
UPS	600-1000 VA	Protects from power cuts and voltage fluctuations; prevents data loss
Networking	Ethernet port + Wi-Fi	Internet, email, online banking, GST filing
Software	Genuine OS, antivirus, office suite	Security and legal compliance

Other points to advise

- Buy from a reputed brand/dealer with **1-3 years warranty** and after-sales service.
- Keep the purchase within the budget; spend more on SSD and RAM rather than on unnecessary graphics.
- Choose hardware that is **upgradable** in future (spare RAM slots, extra SATA/M.2 slots).
- Regular backup of business data and use of antivirus.

Q.1 (c) [10 Marks]

Question: *Compare the hardware components of a desktop computer with those of a smart TV.*

Sl.	Parameter	Desktop Computer	Smart TV
1	Purpose	General-purpose computing (office, programming, design, gaming)	Mainly video/audio entertainment and streaming with limited apps
2	Processor	Powerful x86 CPU (Intel/AMD), replaceable	Low-power ARM-based System-on-Chip (SoC), fixed on board
3	RAM	4-64 GB, user-upgradable DDR modules	1-4 GB, soldered, not upgradable
4	Storage	HDD / SSD (256 GB to many TB), upgradable	Small flash/eMMC memory (8-64 GB) for OS and apps
5	Display	Separate external monitor (LED/LCD)	Built-in large LED/OLED/QLED panel
6	Graphics	Integrated or dedicated GPU card	GPU part of the SoC, video decoding
7	Input devices	Keyboard, mouse (any USB device)	Remote control, voice control, touch not common
8	Operating system	Windows, Linux, macOS	Android TV, Tizen, webOS, Roku OS
9	Connectivity	Ethernet, Wi-Fi, USB, HDMI/DP, audio jacks	HDMI, USB, Wi-Fi, Bluetooth, LAN, antenna/tuner input
10	Audio	External speakers / sound card	Built-in speakers, ARC/optical output
11	Power supply	SMPS inside cabinet (300-650 W)	Internal power board (approx. 50-200 W)
12	Cooling	Fans / heat sinks, active cooling	Mostly passive cooling
13	Upgradability / repair	Every part replaceable	Very limited; board-level repair
14	Special parts	Motherboard, expansion slots, optical drive	TV tuner, panel driver (T-con) board, backlight unit

Conclusion: Both contain processor, memory, storage, display, input and I/O, but a desktop is modular, powerful and upgradable, whereas a smart TV is a compact, fixed, low-power embedded system built for media.

OR

Q.2 (a) [5 Marks]

Question: *Describe the procedure to test an ATX power supply unit using a PSU tester and how the readings help identify a fault.*

PSU tester: A small device with sockets for the PSU connectors and LEDs/LCD that show the output voltage of each rail, so the PSU can be tested without a motherboard.

Procedure

1. Switch OFF the PSU and unplug the AC cable. Disconnect all PSU connectors from the motherboard and drives.
2. Connect the **24-pin main connector** to the tester. Also connect the 4/8-pin CPU, SATA, Molex and PCIe connectors if the tester has sockets.
3. Plug in the AC cable and turn ON the PSU switch at the rear. (If the tester has no auto-start, a PS_ON wire (green) short to ground starts the PSU.)
4. Observe LEDs/LCD: **+3.3 V, +5 V, +12 V, -12 V, +5 VSB** and **PG (Power Good)** signal.
5. Compare the readings with the standard values and tolerance. Note any LED that is off or any reading outside the limit.
6. Switch OFF and unplug the PSU after the test.

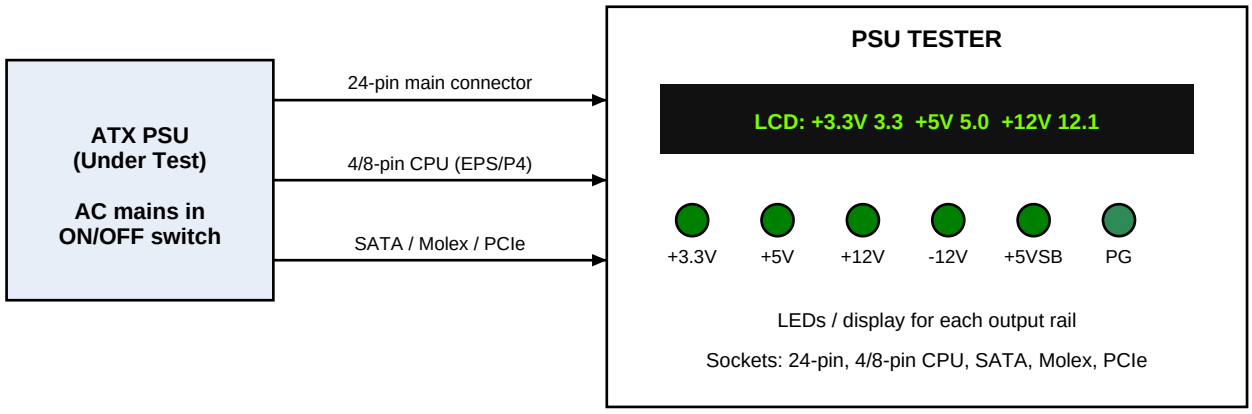

Fig. Testing an ATX PSU with a PSU tester

Standard values and fault identification

Rail	Normal value	Allowed tolerance	If reading is faulty
+3.3 V	3.3 V	+/- 5 % (3.14-3.46 V)	Unstable memory/chipset, random restarts
+5 V	5 V	+/- 5 % (4.75-5.25 V)	Drive and USB problems
+12 V	12 V	+/- 5 % (11.4-12.6 V)	CPU/GPU/fan/HDD motor problem, system shut down under load
-12 V	-12 V	+/- 10 %	Serial/legacy device problems
+5 VSB	5 V	+/- 5 %	Computer does not power on / no standby power
PG signal	Power Good LED ON (delay about 100-500 ms)	-	LED off or blinking: motherboard will not start

- **All LEDs off / no display:** PSU dead, fuse or AC cable faulty.
- **Voltage out of tolerance:** faulty capacitors/regulation circuit; replace PSU.
- **PG off:** PSU defective even though other rails are present.
- All readings within limits: PSU is good; fault lies elsewhere (motherboard, RAM etc.).

Q.2 (b) [10 Marks]

Question: *Divya touched the metal body of an unplugged CPU cabinet after walking on a carpet and felt a spark. Analyse the cause and suggest safety measures against static electricity.*

Analysis of the cause - Electrostatic Discharge (ESD)

1. Walking on a carpet with shoes/synthetic clothes causes **friction (triboelectric effect)**; electrons are transferred and Divya's body gets charged to a high static voltage (can be several thousand volts).
2. The cabinet is unplugged, so it has no mains supply, but its metal body is at a different (nearly earth/neutral) potential.
3. When she touched the metal body, a large **potential difference** existed between her body and the cabinet. Charge flowed suddenly through the contact point, producing a **spark and a small shock** - this is ESD.
4. Dry air (low humidity) increases charge build-up. The spark was felt because humans feel ESD above about 3000 V.
5. Danger to hardware: electronic components (CPU, RAM, chips) can be damaged by ESD as low as **10-100 V**, often without any visible sign. So the spark was not caused by the power supply but by static charge of her body.

Safety measures while handling computer hardware

Sl.	Safety measure
1	Always switch OFF the system and unplug the power cable before opening the cabinet.
2	Wear an **anti-static wrist strap** connected to the cabinet metal/earth ground.
3	Work on an **anti-static mat** and use ESD-safe tools.
4	Touch an earthed metal part (cabinet frame) to discharge the body before touching any component.
5	Hold cards, RAM and boards by their **edges**; never touch chips, pins and gold contacts.
6	Keep components in **anti-static bags** until installation; never keep them on carpet, plastic or cloth.
7	Avoid synthetic clothes, woollen wear and rubber-soled shoes in the work area; avoid carpets - use ESD flooring.
8	Maintain humidity of 40-60 % in the work area using humidifier.
9	Handle one component at a time and avoid unnecessary movement or sliding of components.
10	Use an ionizer in ESD-sensitive labs and follow proper earthing of electrical supply.

Q.2 (c) [10 Marks]

Question: *Explain the working principle of an SMPS with its stages.*

SMPS (Switched Mode Power Supply): An electronic power supply that converts AC mains into regulated DC voltages required by the computer (+3.3 V, +5 V, +12 V etc.) using a **high-frequency switching** (20-100 kHz) transistor and a small transformer. Regulation is done by controlling the ON/OFF time (duty cycle) of the switch using PWM.

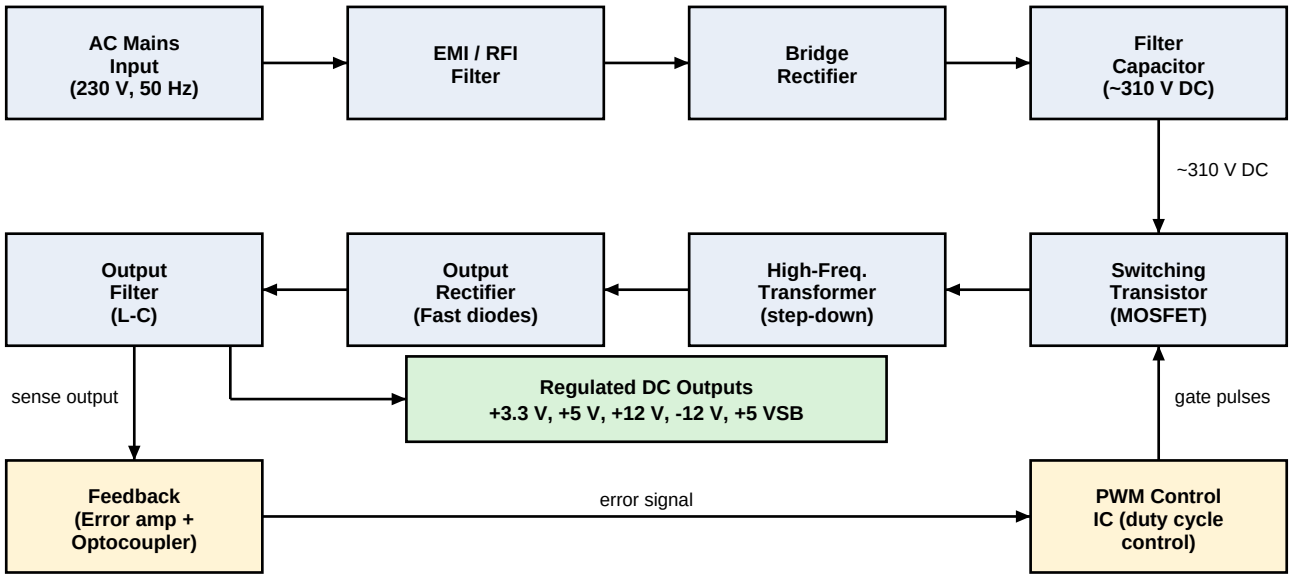

Fig. Block diagram of SMPS (AC to regulated DC)

Output taken from output filter stage; part of it is fed back to PWM controller

Working / Stages

1. **Input stage (EMI filter):** AC mains 230 V, 50 Hz enters through fuse and EMI/RFI filter, which removes noise and surges.
2. **Rectifier:** A bridge rectifier converts AC into pulsating DC.
3. **Filter:** A large capacitor smooths the pulsating DC into unregulated high-voltage DC (about 310 V).
4. **Switching stage (Inverter/chopper):** MOSFET/transistor switches this DC ON and OFF at high frequency (20-100 kHz), converting DC into high-frequency AC/pulses.
5. **High-frequency transformer:** Steps the voltage down and gives electrical isolation from mains. At high frequency the transformer is small and light.
6. **Output rectifier and filter:** Fast-recovery diodes rectify the secondary voltage; L-C filter removes ripple to give smooth DC outputs (+3.3 V, +5 V, +12 V, -12 V, +5 VSB).
7. **Feedback and control:** Output voltage is sensed, compared with a reference by error amplifier and sent through an **optocoupler** to the PWM controller IC. The controller changes the duty cycle of the MOSFET to keep output constant despite load or input changes. It also gives protection (OVP, OCP, short circuit) and the Power Good signal.

Advantages of SMPS

- High efficiency (70-90 %), less heat.
- Small size and light weight.
- Wide input voltage range and good regulation.

SECTION II

Q.3 (a) [5 Marks]

Question: *Nisha and Arjun disagree on which chipset transfers data faster, the north bridge or the south bridge. Compare and conclude.*

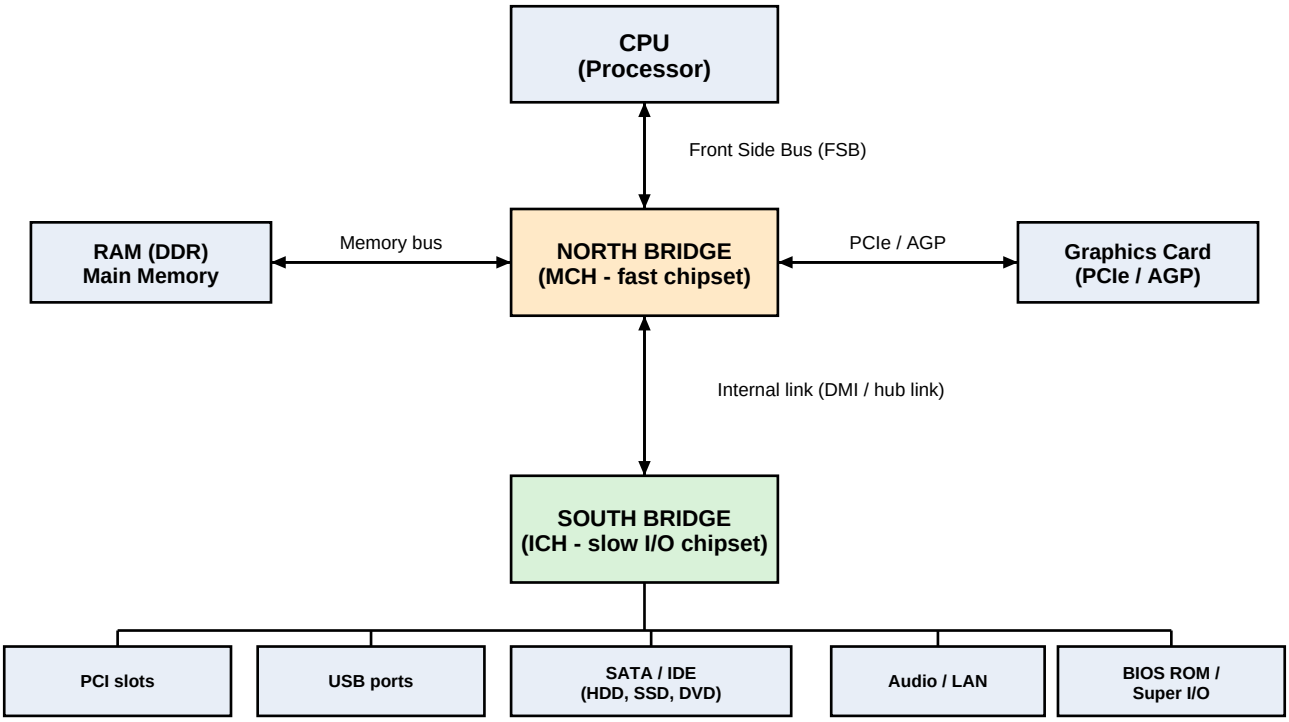

Fig. Motherboard chipset: North bridge and South bridge

Parameter	North Bridge	South Bridge
Connected to	CPU, RAM, graphics card (PCIe/AGP)	Slow I/O: PCI, USB, SATA/IDE, audio, LAN, BIOS
Location	Close to CPU, near top of the board	Near bottom, near PCI slots
Speed	**High speed** (FSB/memory bus, several GB/s)	Lower speed (MB/s)
Link to CPU	Directly through Front Side Bus	Through the north bridge
Heat	Hot, needs heat sink	Less heat
Other names	Memory Controller Hub (MCH)	I/O Controller Hub (ICH)

Conclusion: The **North Bridge transfers data faster**, because it handles high-bandwidth devices (CPU, RAM, graphics) and is directly linked to the CPU, whereas the South Bridge manages slow peripherals. (In modern systems, north bridge functions are built into the CPU and south bridge is called the PCH.)

Q.3 (b) [10 Marks]

Question: *A media player stopped responding on a lab computer; other files are open and unsaved so the system cannot be restarted. How will you close the media player?*

Idea: Do not restart. Terminate only the hung application (process) so the other open programs and unsaved files remain safe.

Method 1: Task Manager (Windows)

1. Press **Ctrl + Shift + Esc** (or Ctrl + Alt + Del and choose Task Manager; or right-click taskbar).

2. Open the **Processes** tab; the hung media player shows status *Not responding*.
3. Select the media player (e.g. VLC / Windows Media Player).
4. Click **End task**. If asked, confirm. The player closes while other programs keep running.
5. Check that the other files are still open; save them immediately.

Method 2: Command line

- Open Command Prompt and type **tasklist** to see running processes.
- Type **taskkill /IM vlc.exe /F** (replace with player's process name) or **taskkill /PID <pid> /F**.

Method 3: Other options

- Press **Alt + F4** on the player window and wait for it to close.
- Linux: use **System Monitor** / **xkill** / **kill <pid>**; Mac: Force Quit (Cmd+Opt+Esc).
- Wait 30-60 seconds first, as heavy video decoding may only be slow.

Precautions / Follow-up

- Save work in the other programs at once. Restart the system later at a convenient time.
- Find the cause: high CPU/RAM usage, corrupted video file, missing codec, outdated player/driver - update or reinstall.

Q.3 (c) [10 Marks]

Question: *Describe the sequence of events from switching on a computer to the loading of the operating system, highlighting the role of BIOS/UEFI and POST.*

The process of starting a computer and loading the OS into RAM is called **booting**.

Sequence of events

1. **Power ON:** Pressing the power button makes the SMPS supply power to the motherboard. After voltages stabilise, it sends the **Power_Good** signal.
2. **CPU reset:** On receiving Power_Good the CPU is reset and starts executing instructions from a fixed address in ROM/flash where the **BIOS/UEFI** is stored.
3. **BIOS/UEFI starts:** It is the firmware that initialises and tests hardware, provides basic input/output services and finds the OS.
4. **POST (Power-On Self Test):** Checks CPU, RAM, video card, keyboard, chipset and drives. If a fault is found, it gives **beep codes** or on-screen error message and stops.
5. **Hardware initialisation:** Settings stored in CMOS (powered by CMOS battery) such as date/time and boot order are read. Devices are configured.
6. **Boot device selection:** Firmware searches devices in the boot order (SSD/HDD, USB, DVD, network).
7. **Boot sector loading:** In BIOS (legacy) mode it reads the **MBR** of the disk; in UEFI mode it loads the bootloader from the **EFI System Partition**.
8. **Bootloader:** Windows Boot Manager or GRUB loads the OS kernel into RAM.
9. **OS loading:** Kernel starts, drivers and services are loaded, and the login screen/desktop appears.

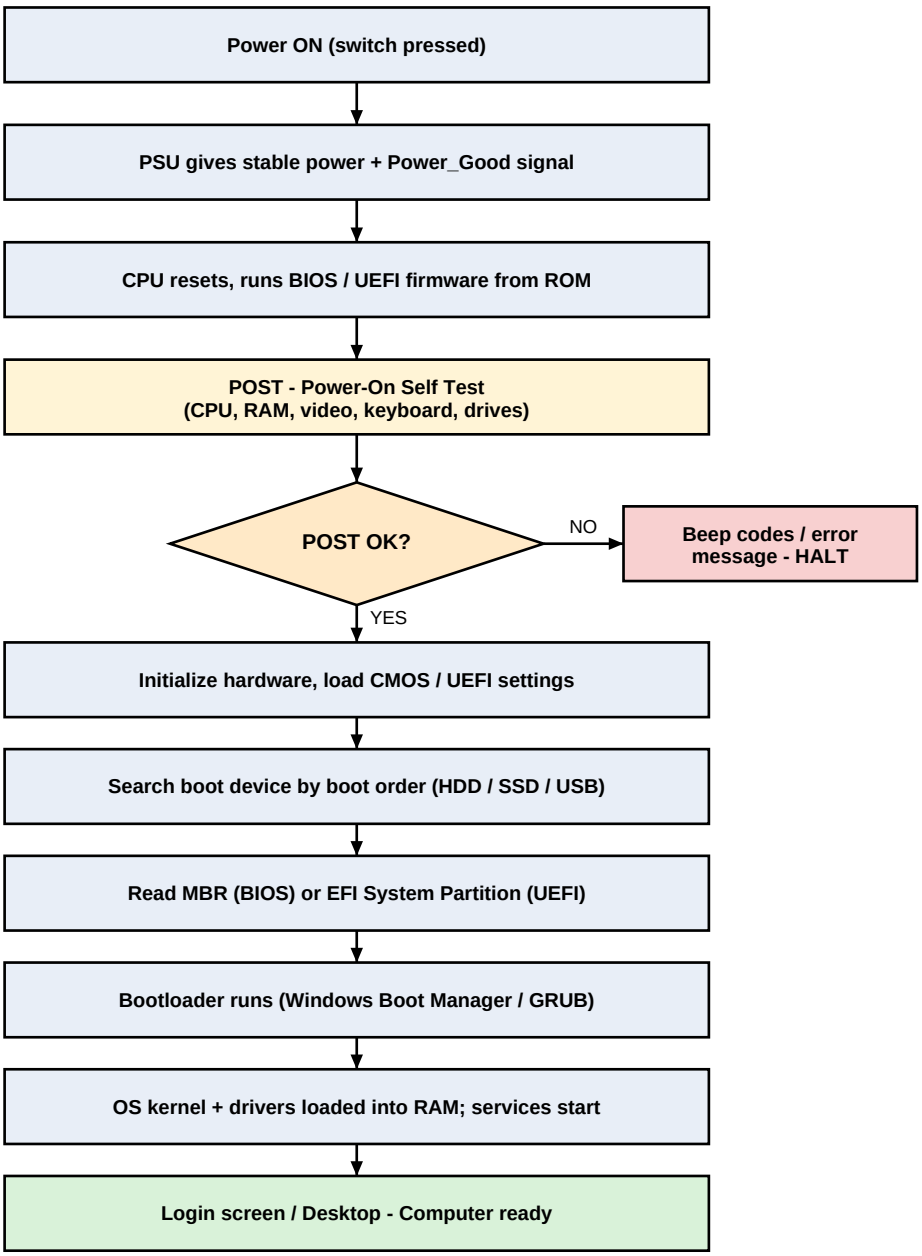

Fig. Boot sequence of a computer

Role of BIOS/UEFI and POST

Item	Role
BIOS/UEFI	Firmware in flash ROM; starts hardware, runs POST, holds setup utility, selects boot device and passes control to bootloader. UEFI is the modern replacement - supports GPT, faster boot, Secure Boot, large disks (>2 TB), mouse/GUI.
POST	Self-test of essential hardware before the OS loads; reports errors by beep codes/messages; prevents booting with faulty hardware.

OR

Q.4 (a) [5 Marks]

Question: *A lab computer gives one long and three short beeps and displays nothing. How can you use POST beep codes to locate the faulty component?*

POST beep codes are audio signals given by the motherboard speaker when POST detects a hardware error before the display is ready. The pattern of long and short beeps identifies the faulty part.

Procedure to locate the fault

1. Note the pattern exactly: **1 long + 3 short beeps**.
2. Identify the **BIOS manufacturer** (AMI, Award, Phoenix) from the motherboard/BIOS label or manual, since the codes differ between makers.
3. Look up the code in the motherboard manual or the manufacturer's beep-code table.
4. Inspect the component the code points to.

Interpretation of 1 long + 3 short

BIOS	Typical meaning
AMI (as commonly listed)	Memory (RAM) failure - conventional/extended memory test
Award (as commonly listed)	Video/graphics card or video memory error

Because the screen shows nothing, the likely faulty parts are the **video card** or **RAM**. (Exact meaning must be confirmed in the manual of the particular board, as it varies.)

Corrective action

- Switch off, unplug and use anti-static precautions.
- Reseat the RAM and video card firmly; clean contacts with an eraser/contact cleaner.
- Try the RAM in another slot / test with a known-good module; test the graphics card in another PC or use onboard video.
- Check monitor cable and monitor. Replace the faulty part and power on again.

Common beep pattern	Usual meaning
No beep / continuous beeps	Power supply or motherboard fault / RAM problem
1 short	POST successful (normal)
Repeated short beeps	Power/RAM problem
Continuous long beeps	Memory fault

Q.4 (b) [10 Marks]

Question: *The accounts office computer, in use for three years, takes long to boot and freezes often. Suggest the hardware and software checks and upgrades to improve performance.*

Possible causes: dust and overheating, ageing HDD, low RAM, too many startup programs, malware, fragmented/full disk, outdated drivers/OS.

A. Hardware checks and upgrades

Check / Upgrade	Action	Benefit
Cleaning and cooling	Clean dust from cabinet, fans, heat sink; replace thermal paste; check fan working	Prevents overheating and thermal throttling
Storage	Check HDD health (S.M.A.R.T.); **replace HDD with SSD**	Boot time reduces greatly; fewer freezes
RAM	Check usage in Task Manager; add RAM (e.g. 4 GB to 8 GB); reseat modules, run memory test	Less swapping, smoother multitasking
Power supply	Test PSU voltages; replace if weak	Stable power, no random freezing
Connections	Check cables, cards and CMOS battery	Removes intermittent faults

Check / Upgrade	Action	Benefit
CPU/Motherboard	Check temperature; upgrade only if very old	Better speed

B. Software checks and optimisation

1. **Scan for virus/malware** with updated antivirus and remove threats.
2. **Disable unnecessary startup programs** (Task Manager > Startup, msconfig).
3. **Uninstall** unused software; remove temporary files with Disk Cleanup; keep at least 15-20 % disk free.
4. Run **chkdsk** to repair disk errors; **defragment** if it is an HDD (not for SSD).
5. Update OS, drivers and BIOS; check Event Viewer for repeated errors.
6. Check running processes for high CPU/memory/disk usage.
7. Take full **backup** of accounts data, and if the problem remains, reinstall/upgrade the OS.

Conclusion: SSD upgrade, extra RAM, cleaning and malware removal together restore the performance of the accounts computer.

Q.4 (c) [10 Marks]

Question: *Differentiate between an SSD and an HDD on the basis of performance, price, reliability and storage capacity.*

Sl.	Basis	SSD (Solid State Drive)	HDD (Hard Disk Drive)
1	Working principle	Stores data in NAND flash memory chips; no moving parts	Stores data magnetically on spinning platters read by moving head
2	**Performance** (speed)	Very fast: 500 MB/s (SATA) to 3500+ MB/s (NVMe); access time about 0.1 ms; boot in seconds	Slow: 80-160 MB/s; access time 5-10 ms; longer boot and load time
3	**Price**	Costlier per GB	Cheaper per GB
4	**Reliability**	No moving parts: shock/vibration resistant, silent; limited write cycles but long life in normal use	Mechanical parts can wear/fail; head crash on shock; more failure-prone
5	**Storage capacity**	Common 128 GB - 2 TB (up to 8 TB, costly)	Common 500 GB - 8 TB (up to 20+ TB)
6	Power consumption	Low	Higher (motor)
7	Noise and heat	Silent, cool	Noisy, more heat
8	Weight and size	Light, small (2.5 inch, M.2)	Heavier (3.5/2.5 inch)
9	Defragmentation	Not needed	Needed periodically
10	Use	Laptops, OS drive, high-performance PCs	Bulk storage, backup, servers

Conclusion: SSD is better in speed and reliability; HDD is better in price per GB and capacity.

Final Check

Item	Status
Every question and sub-question answered (Q1, Q2, Q3, Q4: a, b, c)	Done
Both OR alternatives solved	Done
Marks-wise depth (5 / 10 / 10)	Done
Diagrams: PSU tester (2a), SMPS (2c), North/South bridge (3a), Boot flowchart (3c)	Included and labelled
Tables for classification, comparison and differentiation	Included
Note: beep-code meaning depends on BIOS maker (Q4a)	Stated